

Maverick Espinosa

AI/ML & Data Engineer

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AI/ML and data engineer with an M.S. in Computer Science and experience building LLM evaluation systems, research data pipelines, telemetry tools, and full-stack applications. Combines evidence-driven experimentation with interactive systems that make complex work easier to understand and act on.

EDUCATION

University of Illinois Urbana-Champaign — M.S. in Computer Science, GPA: 3.91/4.0

May 2026

Johns Hopkins University — B.S. in Computer Science, GPA: 3.56/4.0

Dec 2024

SKILLS

AI / ML: PyTorch, Hugging Face, vLLM, LLM agents and evaluation, RAG, GRPO

Languages / Data: Python, TypeScript, JavaScript, R, SQL, pandas, ETL pipelines, data validation

Applications / Systems: React, Next.js, FastAPI, Flask, MongoDB, PostgreSQL, Docker, AWS, Linux, pytest, CI/CD

EXPERIENCE

PactSpace — CTO

Aug 2025 – Aug 2026

- Designed AI-assisted service matchmaking and negotiation workflows for a multi-tenant B2B platform, defining lifecycle state machines, data models, vendor onboarding, and milestone-payment logic
- Interviewed startups and small businesses to map procurement workflows, translating stakeholder needs into prioritized product requirements and implementation plans

Johns Hopkins University Applied Physics Laboratory — Software/Data Engineer Intern

June 2025 – Aug 2025

- Built a full-stack mission-analysis feature with Python, TypeScript, React, and MongoDB, connecting protobuf telemetry to interactive maps, filters, charts, and reports for operational use
- Developed repeatable pipelines to decode, validate, standardize, and ingest autonomous-system telemetry and calculate vessel-stability metrics

Center for Psychedelic and Consciousness Research — Data Analyst / Data Engineer

July 2023 – Aug 2025

- Built R and Python pipelines to extract, clean, score, validate, and harmonize REDCap and Qualtrics research data for analytics and machine-learning workflows
- Developed sensitive-data detection, clinical-data search, and adverse-event review tools that improved data traceability and automated research workflows

WISE Cities — Software Engineer Intern

Sept 2024 – Dec 2024

- Built React, Next.js, tRPC, and MongoDB product features across frontend, API, backend, and localization workflows in a 5-person Agile team
- Instrumented an LLM-powered search workflow with Langfuse and added Sentry observability to support relevance analysis, monitoring, and debugging

PROJECTS

Symbiotic-SWE — Software Repair and Evaluation System

Jan 2026 – May 2026

- Built a Python software-repair system combining LLM-generated patches, program slicing, CrossHair/Z3 verification, counterexample-guided critique, and pytest evaluation
- Developed reproducible SWE-bench workflows for repository setup, patch generation, verification, testing, and run-artifact collection

Molecule Generation with Reinforcement Learning

Jan 2026 – May 2026

- Implemented a GRPO-style post-training loop for DiffDock with grouped pose generation, RMSD rewards, native losses, checkpoint export, and matched evaluation
- Tested 2 preliminary local seeds; 1 improved and the second did not reproduce the result, identifying update stability as the central limitation